

Beyond Classical Cryptography: Simulating & Benchmarking Quantum Key Distribution

Om Kalyan

230733440

A dissertation submitted in partial fulfilment of the requirements
for the degree of

Bachelor of Engineering
Electronics & Computer Engineering



School of Engineering
Newcastle University

2026

ABSTRACT

Quantum algorithms such as Shor's factorisation threaten public-key cryptography systems like RSA and ECC, which rely on computational hardness for security. Quantum Key Distribution (QKD) offers a fundamentally different security paradigm: rather than relying on computational complexity, QKD derives information-theoretic security from the laws of quantum mechanics, ensuring privacy regardless of adversarial computing power. This project presents a simulation and benchmarking platform implementing three QKD protocols: BB84, B92, and E91 in Python3 using the Qiskit SDK. The platform compares Prepare & Measure approaches against Entanglement-Based key distribution, evaluating protocol performance first under ideal channel conditions to establish theoretical baselines, then under realistic constraints including eavesdropper presence, quantum bit error rates, and channel losses.

ACKNOWLEDGEMENTS

I would like to thank my supervisors Martin Johnston and Oktay Çetinkaya for their guidance. . .

CONTENTS

Abstract	i
Acknowledgements	ii
List of Figures	v
List of Tables	vi
List of Listings	vii
List of Algorithms	viii
1 Introduction	1
I A Brief History of Cryptology	1
II The Quantum Threat to Classical Cryptography	1
III Aims & Objectives	2
A Literature Review & Analytical Framework	2
B Protocol Implementation	2
C Channel Modelling and Eavesdropper Simulation	2
D Post-Processing and Secure Key Rate Analysis	2
E Verification, Validation, and Benchmarking Visualisation	2
F Real-World Feasibility and Broader Context	3
2 Literature Review	4
I The Quantum Threat to Classical Cryptography	4
II Quantum Mechanical Foundations	5
A Qubits & Superposition	5
B Measurement Disturbance	5
C No-cloning Theorem	5
D Heisenburg’s Uncertainty	6
E Quantum Entanglement & Bell’s Theorem	6
III Quantum Key Distribution Protocols	7
A BB84	7
B B92	8
C E91	9
IV The Post-Processing Pipeline	9
V Noise Models for Modelling	9
VI QKD Simulation Frameworks	9
VII Research Gaps	9
3 Methodology	10
I Simulation Architecture	10
II Protocol Implementations	10
A BB84 Protocol	10
1 Basis Selection	10
B B92 Protocol	10
C E91 Protocol	10
III Modelling Assumptions	10
IV Noise Models and Channel Simulation	10
V Performance Metrics	10
4 Results & Discussion	11
I Simulation Outputs	11

II	BB84 vs B92	11
III	Performance Validation	11
IV	Implications	11
5	Conclusion & Future Work	12
A	Source Code	15
B	Additional Results	16

LIST OF FIGURES

2.1	The Quantum Economic Advantage	4
2.2	Classification of QKD Protocol Schemes	7
2.3	Abstract P&M QKD Network Model	7
2.4	BB84 Operation Model	8
2.5	B92 Measurements Bases	8

LIST OF TABLES

LIST OF LISTINGS

3.1	BB84 Bob's measurement logic	10
-----	--	----

LIST OF ALGORITHMS

1	BB84 Key Sifting Procedure	10
---	--------------------------------------	----

CHAPTER 1

INTRODUCTION

I. A BRIEF HISTORY OF CRYPTOLOGY

Since the dawn of days, humans have sought ways to communicate in secret. It only takes a glance back at the ancient societies of Mesopotamia, India, and China to understand where the foundations of cryptology were laid some 3000 years ago [1]. Whether it be the early use of Substitution or Transposition schemes, these early cryptographic systems established the core principle that would dominate cryptology for millennia: security through the obscurity of the method itself.

Back in the 9th century, humanity experienced the first major analytical breakthrough when the Arab polymath Al-Kindi developed frequency analysis, a technique of cryptanalysis capable of breaking any monoalphabetic substitution cipher of the age by exploiting the statistical patterns of language itself [2]. This initiated a centuries-long cycle of invention and defeat that defined the field of cryptology, with newly designed ciphers broken by increasingly sophisticated cryptanalysis. Fast-forward to the 16th century, Blaise de Vigenère built on the work of Giovan Bellaso to introduce a polyalphabetic cipher that would go on to secure the world for 300 years, until the great minds of Babbage and Kasiski developed statistical methods to compromise the scheme in the 19th century [3, 4].

The dawn of the Second World War transformed cryptology from a niche discipline into a strategic imperative. Repeating this "Tug of War", Germany's Enigma, a rotor-based electromechanical cipher, was famously broken at Bletchley Park by the Father of Theoretical Computer Science - Alan Turing, and his colleagues [5]. Despite this technological revolution, security through simplicity was demonstrated in the Pacific theatre by the US Marine Corps' deployment of Navajo Code Talkers who used their native language as the basis for a battlefield cipher. A truly innovative system that rested security on the extreme phonetic distance between the Navajo and Japanese languages remained unbroken until its declassification in 1968 [6].

Looking to the post-war era, Claude Shannon redefined cryptography using rigorous mathematical foundations in his initially classified 1945 paper on secrecy systems, introducing concepts such as confusion, diffusion, and perfect secrecy [7]. This revolutionary framework, later publicly published in 1948, fueled the development of the modern cryptographic infrastructure we rely on today: the Data Encryption Standard (DES) in 1977, the RSA public-key cryptosystem in 1978, and the Advanced Encryption Standard (AES) in 2001 [8–10]. Peeling back the layers of essentially all the digital communications we use today, from online banking and e-commerce to government and military networks, we find these algorithms underpinning the security on which we rely.

II. THE QUANTUM THREAT TO CLASSICAL CRYPTOGRAPHY

The security of modern public-key cryptography rests on the computational hardness of problems like integer factorisation and the discrete logarithm. Shor's algorithm [11] demonstrated that a sufficiently powerful quantum computer could solve both in polynomial time, rendering schemes such as RSA and ECC fundamentally insecure. Two complementary responses have emerged: Post-Quantum Cryptography (PQC), which replaces vulnerable algorithms with quantum-resistant mathematical problems, and Quantum Key Distribution (QKD), which exploits the laws of quantum mechanics to distribute encryption keys with information-theoretic security. This project focuses on the latter, building a simulation and benchmarking platform for QKD protocols including BB84, B92, and E91.

III. AIMS & OBJECTIVES

The aim of this project is to design, implement, and evaluate a modular simulation and benchmarking platform for Discrete-Variable Quantum Key Distribution protocols, enabling quantitative comparison of protocol performance under realistic channel conditions, eavesdropping attacks, and post-processing constraints. All simulations are conducted in Python using IBM's Qiskit framework and the Aer noise simulator. The specific objectives are organised into six categories.

A. Literature Review & Analytical Framework

- O1.1** Conduct a critical review of QKD protocol literature, covering the theoretical foundations of BB84 [12], B92 [13], and E91 [14], their respective security proofs, and known practical limitations. This review will establish the analytical framework and parameter choices (noise models, error correction efficiency, security bounds) used throughout the project.
- O1.2** Identify and document the key metrics by which QKD protocols are evaluated in the literature, such as: Quantum Bit Error Rate (QBER), sifted key rate, secure key rate, maximum tolerable error threshold, and sifting efficiency. These metrics will form the basis of all subsequent benchmarking.

B. Protocol Implementation

- O2.1 BB84 (Four-State, Two-Basis):** Implement the BB84 protocol as the primary baseline simulation. The implementation models Alice's random bit and basis selection, qubit state preparation in the rectilinear (Z) or diagonal (X) basis, quantum transmission through a configurable noisy channel, Bob's random basis measurement, and public basis reconciliation (sifting). The implementation is considered complete when the sifting efficiency converges to the theoretically expected $\sim 50\%$ across repeated simulation runs.
- O2.2 B92 (Two Non-Orthogonal States):** Implement Bennett's 1992 protocol as the primary comparative prepare and measure protocol, encoding bit values using only two non-orthogonal states ($|0\rangle$ and $|+\rangle$). Bob performs a measurement that can yield a conclusive result, an inconclusive result, or a photon loss event. The implementation is validated by confirming that the sifting efficiency converges to $\sim 25\%$ and that the QBER threshold falls below that of BB84, consistent with B92's reduced noise tolerance.
- O2.3 E91 (Entanglement-Based):** Implement Ekert's 1991 protocol. This models the generation and distribution of maximally entangled Bell pairs, independent random basis measurements from three orientations per party, key extraction from matching-basis outcomes, and eavesdropper detection via the CHSH inequality. The implementation is validated by confirming that the computed CHSH parameter $|S|$ converges to $2\sqrt{2} \approx 2.828$ in the absence of eavesdropping, and degrades towards the classical bound of 2 under increasing interception rate.

C. Channel Modelling and Eavesdropper Simulation

- O3.1** Model quantum channel effects using Qiskit Aer's noise framework: depolarising noise, and phase damping. Noise is applied exclusively to the quantum channel via an identity gate channel marker, not to local preparation or measurement operations, ensuring the simulation accurately represents a protocol where Alice's and Bob's local devices are trusted.
- O3.2** Implement an intercept-resend eavesdropping attack with a configurable interception probability $p \in [0, 1]$. Eve intercepts each qubit with probability p , measures in a randomly chosen basis, and re-transmits to Bob. Sweep the interception rate from 0% to 100% record the resulting QBER for each protocol.

D. Post-Processing and Secure Key Rate Analysis

- O4.1** Calculate the secure key rate using the GLLP formula [15], incorporating the error correction efficiency factor $f_{EC} = 1.16$ corresponding to the CASCADE protocol [16]. This yields a practical BB84 QBER security threshold of $\sim 9.8\%$, compared to the idealised 11% Shor-Preskill bound at $f_{EC} = 1.0$ [17]. Document and justify this distinction, as it has direct engineering significance for determining whether a given fibre link can sustain secure key generation.
- O4.2** Compute the mutual information quantities $I(A:B)$ (Alice-Bob) and $I(A:E)$ (Alice-Eve) as functions of QBER for both BB84 and B92. Identify the crossover point where $I(A:E) \geq I(A:B)$, beyond which no secret key can be extracted, and confirm this aligns with the GLLP-derived thresholds for each protocol.
- O4.3** Determine and compare the maximum tolerable QBER thresholds for BB84 ($\sim 9.8\%$ with $f_{EC} = 1.16$) and B92 ($\sim 6.5\%$), quantifying the practical noise tolerance gap between the two prepare and measure protocols.

E. Verification, Validation, and Benchmarking Visualisation

- O5.1 Verification against theoretical predictions.** Confirm that simulation outputs match known analytical results:
 - BB84 sifting efficiency converges to 50% ($\pm 2\%$).
 - Full intercept-resend attack on BB84 produces QBER of 25% ($\pm 2\%$).

- Secure key rate reaches zero at the GLLP-predicted QBER thresholds for each protocol.
- B92 sifting efficiency converges to 25%.

These checks serve as pass/fail criteria confirming that the simulation correctly implements the underlying quantum mechanics and post-processing mathematics.

O5.2 Validation against published experimental data. Compare simulated key rate versus distance curves against published experimental QKD demonstrations, including long-distance BB84 results [18] and metropolitan network deployments reported in recent survey literature [19]. The simulation is not expected to reproduce experimental results exactly (real systems include imperfections beyond the scope of this model), but the trends, orders of magnitude, and maximum operational distances should be broadly consistent. Discrepancies are documented and attributed to specific modelling assumptions.

O5.3 Benchmarking visualisation suite. Develop a set of figures to present simulation results and protocol comparisons, including:

- QBER versus eavesdropper interception rate for each protocol.
- Secure key rate versus fibre distance under realistic attenuation and $f_{EC} = 1.16$.
- Noise model comparison showing QBER response across depolarising, bit-flip, amplitude damping, and phase damping channels at varying strengths.
- Mutual information crossover plot ($I(A:B)$ vs. $I(A:E)$ vs. QBER).

F. Real-World Feasibility and Broader Context

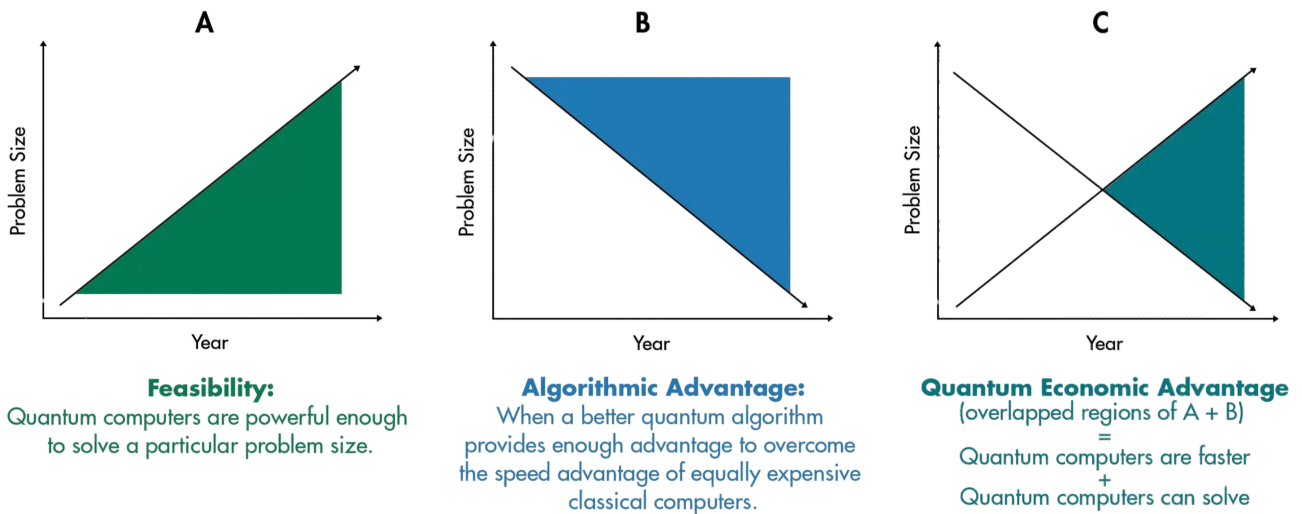
- O6.1** Evaluate the practical viability of BB84 and B92 over metropolitan fibre links (10-50 km), identifying the distance and noise regimes in which each protocol sustains a positive secure key rate with $f_{EC} = 1.16$. Determine, for a given noise floor, the maximum deployment distance at which each protocol remains operational.
- O6.2** Contextualise the simulation results within the broader landscape of quantum-safe security, including the complementary roles of QKD and post-quantum cryptography (NIST FIPS 203/204/205), the "harvest now, decrypt later" threat model motivating near-term QKD deployment, and the economic and infrastructure barriers to adoption across sectors such as Research & Development, Government, Banking & Finance, Critical Infrastructure, Healthcare, and Emerging Industries.

CHAPTER 2

LITERATURE REVIEW

I. THE QUANTUM THREAT TO CLASSICAL CRYPTOGRAPHY

The security of public-key cryptosystems such as RSA, Diffie-Hellman, and ECC depends on the assumed intractability of integer factorisation and the discrete logarithm problem. Shor's algorithm [11] undermines both assumptions, solving each in polynomial time on a quantum computer. While no machine of sufficient scale exists today, resource estimates have fallen sharply. Gidney and Ekerå [20] showed that a 2048-bit RSA modulus could be factored in approximately eight hours using 20 million noisy qubits, and a 2025 follow-up by Gidney [21] reduced the requirement to under one million noisy qubits at the cost of roughly one week of runtime. As illustrated in Fig. 2.1, these declining estimates suggest we are approaching Quantum Economic Advantage (QEA): the point at which the practical resource cost of such an attack becomes feasible.



Credit: "The Quantum Tortoise and the Classical Hare: A Simple Framework for Understanding Which Problems Quantum Computing Will Accelerate (and Which It Won't)." Sukwoong Choi, William S. Moses, and Neil Thompson.



Fig. 2.1. The Quantum Economic Advantage

Symmetric-key cryptography is not immune. Grover's search algorithm [22] provides a quadratic speed-up over brute-force key search, effectively halving the security level of schemes such as AES-128 and AES-256 [23]. In practice, this means AES-128 offers only 64 bits of security against a quantum adversary, making AES-256 the minimum acceptable key length in a post-quantum setting. The asymmetry of impact is notable, where Shor's algorithm breaks public-key schemes outright, Grover's merely weakens symmetric schemes to a degree that can be compensated by doubling key sizes. It is for this reason that the quantum threat is primarily directed at public-key infrastructure.

The "harvest now, decrypt later" attack model gives these theoretical vulnerabilities immediate practical relevance. An adversary can intercept and store classically encrypted traffic today with the intent of decrypting it once a capable QC becomes available [24]. For data with long confidentiality requirements, such as state secrets, medical records, and financial instruments, the effective window of vulnerability is already open. This threat model does not require a quantum computer to exist now; it simply requires that one will exist within the data's confidentiality lifetime.

Two complementary approaches address this threat. PQC replaces vulnerable mathematical assumptions with problems believed to resist quantum attack. NIST finalised its first three PQC standards in August 2024 (FIPS 203, 204, 205), with a fourth algorithm selected in March 2025 [25]. The accompanying IR-8547 transition roadmap deprecates quantum-vulnerable algorithms by 2030 and disallows them entirely by 2035.

However, as PQC provides *computational* security, its guarantees only hold as long as the underlying mathematical problems remain hard, an assumption that cannot be proven and could be invalidated by future algorithmic breakthroughs. QKD takes a fundamentally different approach. By encoding key material in quantum states and exploiting physical principles such as the no-cloning theorem and measurement disturbance, QKD achieves *information-theoretic* security, remaining secure regardless of an

adversary's computational power, now or in the future [17, 26].

Shannon [7] established that information-theoretic secrecy requires a key at least as long as the message, a condition met by the one-time pad but impractical without a secure means of key distribution. QKD is the only known method of generating and distributing such keys at scale without relying on a trusted courier [23]. Any eavesdropping attempt on the quantum channel inevitably disturbs the transmitted states, producing measurable errors that allow the communicating parties to detect interception and discard compromised key material. This physical guarantee is what distinguishes QKD from all classical and post-quantum alternatives. The role of QKD is therefore not as a competitor to PQC, but as a complementary layer offering a stronger class of security guarantee for applications where long-term confidentiality is critical.

II. QUANTUM MECHANICAL FOUNDATIONS

To achieve information-theoretic security, every QKD protocol rests on a small set of quantum mechanical principles. This section introduces the five principles that are exploited by the BB84, B92, and E91 protocols. The treatment of these foundations follows Nielsen and Chuang [23], and the pedagogical review by [26].

A. Qubits & Superposition

Consider a classical bit taking one of two definite values, 0 or 1. A Qubit (quantum bit) can exist in a linear combination of both 0 or 1, simultaneously. Using Dirac notation, the general single-qubit state is written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (2.1)$$

where α and β are complex probability amplitudes that satisfy $|\alpha|^2 + |\beta|^2 = 1$ [23]. The two states, $|0\rangle$ and $|1\rangle$ form the *computational basis* (a.k.a. rectilinear or Z basis). Another valid description of a qubit uses the diagonal basis (a.k.a X Basis), defined as

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (2.2)$$

In a physical QKD system, qubits are typically encoded in the polarisation of single photons: $|0\rangle$ and $|1\rangle$ may correspond to horizontal and vertical polarisation, while $|+\rangle$ and $|-\rangle$ correspond to $+45^\circ$ and -45° diagonal polarisation [12]. Superposition is what allows a single photon to carry information in more than one basis at once, this property becomes a security asset because any attempt to read the qubit forces a basis choice that irreversibly disturbs the state.

B. Measurement Disturbance

Measuring a qubit that is in superposition collapses it probabilistically into one of the basis states of the chosen measurement apparatus. If Alice prepares $|+\rangle$ (encoding a logical 1 in BB84's diagonal basis) and Bob measures in the matching X basis, he recovers the correct value with certainty. However, if Bob instead measures in the Z basis, he obtains $|0\rangle$ or $|1\rangle$, each with probability $\frac{1}{2}$, and the original superposition is destroyed [23]. The Z and X bases are called *conjugate bases* because perfect knowledge in one precludes any information about the other. This is the physical mechanism behind the sifting step in BB84 and B92: only the rounds where Alice and Bob happen to use compatible bases contribute to the key, while mismatched rounds are discarded because Bob's result is uncorrelated with Alice's bit [12, 13].

Critically, measurement disturbance also applies to an eavesdropper (Eve). If Eve intercepts a qubit and measures it in the wrong basis, she collapses the state and re-sends a qubit that no longer matches Alice's preparation. When Alice and Bob later compare a random subset of their sifted key, they detect these errors as an elevated Quantum Bit Error Rate (QBER). An intercept-resend attack on BB84 introduces a characteristic QBER of approximately 25% [17, 26].

C. No-cloning Theorem

A natural question is whether Eve could copy the qubit in transit, keep the copy for later analysis, and forward the original undisturbed qubit. The no-cloning theorem, proven independently by Wootters & Zurek [27] and by Dieks [28], rules this attack out. The theorem states that no physical process can produce a perfect copy of an arbitrary unknown quantum state. Formally, there exists no unitary operator U such that $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$, for all $|\psi\rangle$ [23]. The proof is compact: suppose such a U did exist for two non-orthogonal states $|\psi\rangle$ and $|\phi\rangle$. Unitarity requires preservation of inner products, giving $\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2$, which is satisfied only when $\langle\psi|\phi\rangle = 0$ or 1. Since $|\psi\rangle$ and $|\phi\rangle$ are non-orthogonal by assumption, neither condition holds, producing a contradiction.

No-cloning is the single most important result underpinning QKD. It guarantees that Eve cannot silently duplicate quantum transmissions and is the reason quantum key distribution can offer information-theoretic security rather than the computational security of classical schemes [29]. B92 exploits this principle particularly directly, where the protocol encodes bits using only two

non-orthogonal states ($|0\rangle$ and $|+\rangle$), no measurement or cloning strategy can distinguish them with certainty [13].

D. Heisenberg's Uncertainty

Closely related to measurement disturbance is Heisenberg's uncertainty principle [30]. In its information-theoretic form relevant to QKD, the principle states that for two non-commuting observables (such as polarisation measured in the Z and X bases), simultaneously obtaining precise values for both is impossible. Any measurement that yields information about one observable necessarily introduces uncertainty in the other [23]. In the context of QKD, this means Eve faces a fundamental trade-off: gaining information about Alice's bit in one basis unavoidably disturbs the conjugate-basis encoding, which manifests as detectable errors. Heisenberg's principle and the no-cloning theorem are logically distinct but operationally complementary; together they establish that any eavesdropping strategy, whether based on measurement, cloning, or a combination of both, will leave a detectable footprint in the quantum channel [26].

E. Quantum Entanglement & Bell's Theorem

Two or more qubits can exist in an *entangled* state that cannot be described as a product of individual qubit states. The canonical example is the Bell state (also called an EPR pair after Einstein, Podolsky and Rosen [31]):

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (2.3)$$

If Alice holds the first qubit and Bob the second, a measurement by Alice in the Z basis yielding $|0\rangle$ instantaneously projects Bob's qubit into $|0\rangle$ as well, regardless of the spatial separation between them [23]. Entanglement does not permit faster-than-light communication (the individual outcomes are still uniformly random), but it does produce correlations that cannot be explained by any local hidden-variable theory.

Bell [32] formalised this distinction in 1964 by deriving an inequality that any local realist theory must satisfy. The experimentally testable form due to Clauser, Horne, Shimony and Holt [33] (the CHSH inequality) states

$$|S| = |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2, \quad (2.4)$$

where $E(a,b)$ denotes the correlation of measurement outcomes for settings a and b . Quantum mechanics predicts a maximum value of $|S| = 2\sqrt{2} \approx 2.83$, a violation confirmed experimentally by Aspect [34].

Ekert's E91 protocol [14] directly leverages Bell inequality violation as a built-in eavesdropping test. While BB84 and B92 do not require entanglement, the Shor-Preskill security proof [17] establishes a formal equivalence between BB84 and an entanglement-based scheme, meaning the security of prepare-and-measure protocols ultimately derives from the same non-local correlations.

III. QUANTUM KEY DISTRIBUTION PROTOCOLS

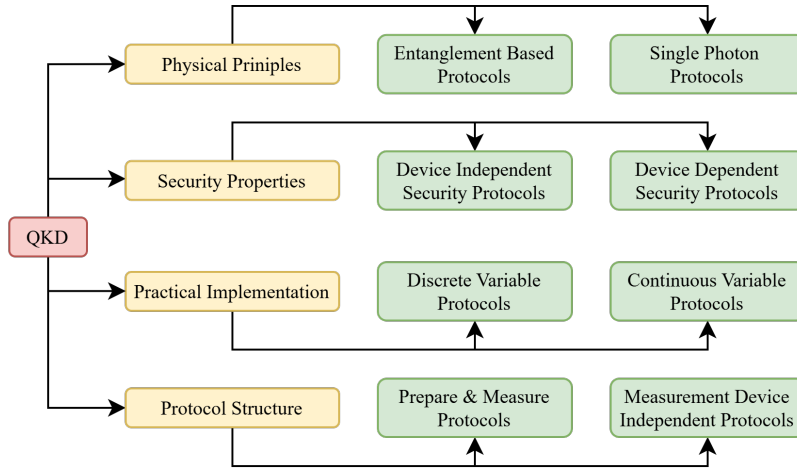


Fig. 2.2. Classification of QKD Protocol Schemes

QKD Protocols can be classified into four basic aspects of use, as depicted in amber in Fig. 2.2. Each aspect is then further classified based upon the quantum mechanical principles exploited, seen in green. Continuous Variable (CV) schemes encode data using the continuous variable of a quantum system, such as the amplitude or phase of a light wave. Due to this, CV protocols can be used with common optical equipment and, as such, are less susceptible to noise when deployed practically. In comparison, Discrete Variable (DV) schemes are implemented using discrete quantum states, such as single polarized photons or entangled-pairs of photons [35]. Furthermore, QKD protocol structures can be classified into three broad paradigms: Prepare & Measure, Entanglement-Based, or Measurement-Device-Independent schemes. This project focuses on three DV-Protocols: BB84, B92 P&M protocols, and the Entanglement-Based E91 protocol.

A. BB84

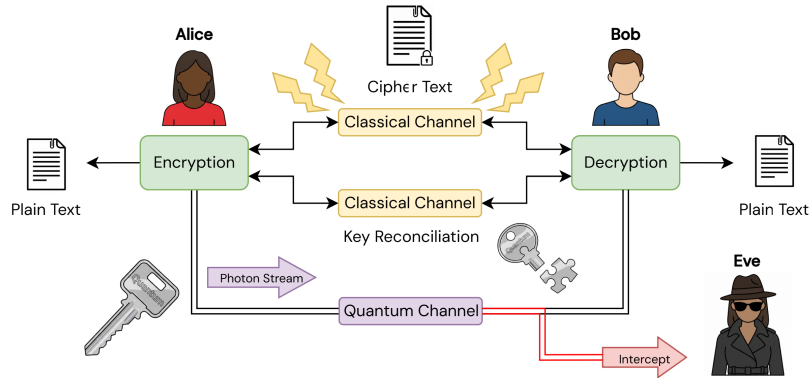


Fig. 2.3. Abstract P&M QKD Network Model

The BB84 protocol, proposed by Bennett and Brassard in 1984 [12], was the first practical QKD scheme. It is a prepare-and-measure protocol that encodes classical bit values onto single-photon polarisation states drawn from two mutually unbiased bases: the rectilinear basis ($|0\rangle, |1\rangle$) and the diagonal basis ($|+\rangle, |-\rangle$). The sender, Alice, randomly selects both a bit value and a basis for each qubit, producing one of four possible states:

$$|0\rangle, |1\rangle, |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (2.5)$$

These states are transmitted to Bob over the quantum channel. Bob then independently and randomly chooses a measurement basis for each incoming qubit. Because the two bases are mutually unbiased, a measurement in the wrong basis yields a uniformly random outcome and provides no information about Alice's bit value [23].

After transmission is complete, Alice and Bob switch to an authenticated classical channel to perform *sifting*, as seen in Fig. 2.3. At this point, Bob announces only his choice of basis for each qubit, and Alice confirms which choices matched her preparation basis. All rounds in which the bases disagree are discarded. In an ideal, noiseless channel, the remaining *sifted key* is perfectly correlated between Alice and Bob, and roughly half the transmitted qubits survive the sifting step [12].

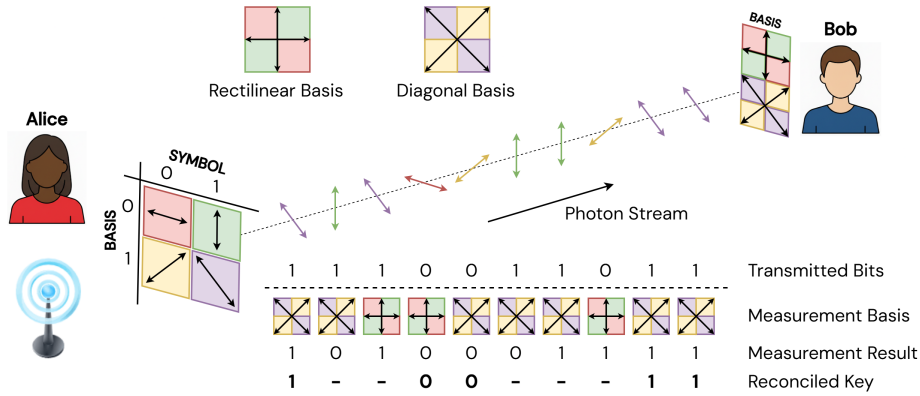


Fig. 2.4. BB84 Operation Model

To check for eavesdropping, Alice and Bob sacrifice a random subset of the sifted key and compare bit values over the public channel. Any intercept-resend attack by an eavesdropper, Eve, necessarily disturbs a fraction of the qubits because she cannot simultaneously determine the polarisation in both bases without violating the no-cloning theorem [23]. A full intercept-resend strategy is expected to produce a Quantum Bit Error Rate (QBER) of approximately 25%, well above the theoretical security threshold of approximately 11% [29]. If the observed QBER exceeds this threshold, the protocol aborts. Otherwise, Alice and Bob proceed through error correction and privacy amplification to yield a final secret key, depicted in Fig. 2.4.

B. B92

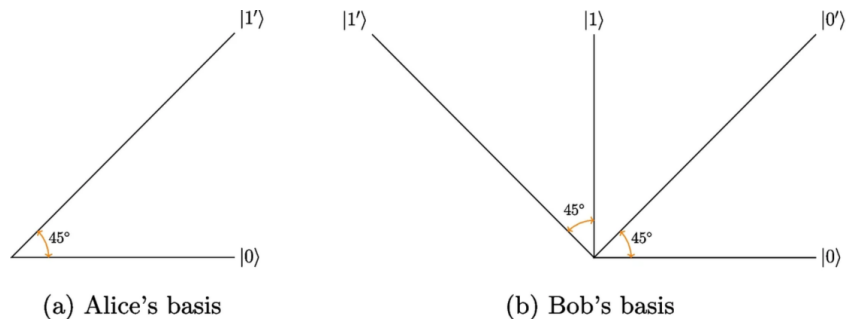


Fig. 2.5. B92 Measurements Bases

In 1992, Bennett proposed the B92 protocol as a simplification of BB84 [13]. The central insight is that unconditional security does not require four encoding states; two non-orthogonal states (seen Fig. 2.5) are sufficient because no measurement can perfectly distinguish between non-orthogonal quantum states [29]. In B92, Alice encodes bit value 0 as $|0\rangle$ and bit value 1 as $|+\rangle$:

$$|\varphi\rangle = \begin{cases} |0\rangle & \text{if } a = 0, \\ |+\rangle & \text{if } a = 1. \end{cases} \quad (2.6)$$

Bob generates his own random bit a' and measures in the Z basis ($|0\rangle, |1\rangle$) if $a' = 0$, or in the X basis ($|+\rangle, |-\rangle$) if $a' = 1$. When Bob's measurement basis is incompatible with Alice's state, the outcome is conclusive only some of the time, hence, inconclusive

results (a.k.a. erasures) are discarded [23]. Bob publicly announces which rounds yielded a conclusive result, and Alice retains only the corresponding bits. Neither party reveals their basis choice or bit value during this exchange.

Because B92 relies on only two states rather than four, its sifting efficiency is lower than that of BB84, and its tolerance to channel noise is reduced. The theoretical QBER security threshold for B92 is approximately 6.5%, compared to approximately 11% for BB84 [29]. Furthermore, the two-state encoding makes B92 more susceptible to unambiguous state discrimination (USD) attacks, in which Eve exploits the known non-orthogonal structure to occasionally identify Alice’s state with certainty while flagging ambiguous cases as losses [13]. Despite these limitations, B92 remains an important protocol both for its theoretical elegance and because it demonstrates that the minimal resource needed for secure QKD is a pair of non-orthogonal states.

C. E91

In contrast to the P&M paradigm of BB84 and B92, the E91 protocol proposed by Ekert in 1991 [14] derives its security from quantum entanglement and Bell’s theorem. A source prepares pairs of qubits in the singlet Bell state:

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \quad (2.7)$$

and distributes one qubit from each pair to Alice and the other to Bob. Each party independently selects a random measurement orientation from a predetermined set of angles. When Alice and Bob happen to choose the same measurement basis, their outcomes are perfectly anti-correlated, and these matching rounds are used to construct the raw key [14].

The rounds in which Alice and Bob chose different bases serve a separate purpose: they are used to evaluate the Clauser-Horne-Shimony-Holt (CHSH) inequality [23]. For the singlet state, quantum mechanics predicts a CHSH parameter of $S = 2\sqrt{2}$, which maximally violates the classical bound of $S \leq 2$. Any eavesdropping that disturbs the entanglement between the pairs will reduce the observed value of S towards or below the classical limit, thereby revealing Eve’s presence [14, 29]. This connection between eavesdropper detection and Bell inequality violation is the defining feature of E91 and distinguishes it fundamentally from P&M protocols.

This entanglement-based approach offers a conceptual advantage as the key bits are not determined until the moment of measurement, so there is no point during transmission at which an encoded classical value exists for Eve to intercept. However, E91 places greater demands on hardware because it requires a source of high-fidelity entangled pairs and the ability to distribute them over the quantum channel while preserving entanglement [29]. In a simulation context, this translates to the additional complexity of modelling entangled state preparation, Bell-state measurement, and CHSH parameter estimation alongside the standard post-processing pipeline shared with BB84 and B92.

IV. THE POST-PROCESSING PIPELINE

V. NOISE MODELS FOR MODELLING

VI. QKD SIMULATION FRAMEWORKS

VII. RESEARCH GAPS

CHAPTER 3

METHODOLOGY

I. SIMULATION ARCHITECTURE

II. PROTOCOL IMPLEMENTATIONS

A. BB84 Protocol

1) *Basis Selection*: The sender (Alice) randomly selects between the rectilinear and diagonal bases for each qubit transmission.

a) *Rectilinear basis*: Uses the computational basis states $|0\rangle$ and $|1\rangle$ for encoding bit values 0 and 1 respectively.

b) *Diagonal basis*: Uses the Hadamard-rotated states $|+\rangle$ and $|-\rangle$ for encoding.

Listing 3.1 shows the measurement logic used by Bob in the BB84 protocol implementation.

```

1 def measure_qubits(qubits, bases):
2     """
3     Measurement logic for QKD receiver.
4     Applies rectilinear or diagonal measurement
5     depending on the randomly chosen basis.
6     """
7     results = []
8     for q, b in zip(qubits, bases):
9         if b == 0:
10             results.append(measure_rectilinear(q))
11         else:
12             results.append(measure_diagonal(q))
13     return results

```

Listing 3.1. BB84 Bob's measurement logic

The high-level flow of the BB84 key sifting procedure is presented in Algorithm 1.

Algorithm 1: BB84 Key Sifting Procedure

Input: Alice's bits a_i , bases $\hat{a}_i$; Bob's measurements b_i , bases $\hat{b}_i$

Output: Sifted key K

```

1  $K \leftarrow \emptyset$ ;
2 for  $i \leftarrow 1$  to  $n$  do
3     Alice publishes  $\hat{a}_i$  over classical channel;
4     Bob publishes  $\hat{b}_i$  over classical channel;
5     if  $\hat{a}_i = \hat{b}_i$  then
6         Append  $b_i$  to  $K$  ▷ Matching basis -- keep bit
7 return  $K$ 

```

B. B92 Protocol

C. E91 Protocol

III. MODELLING ASSUMPTIONS

IV. NOISE MODELS AND CHANNEL SIMULATION

V. PERFORMANCE METRICS

The asymptotic secure key rate under the Shor–Preskill bound is given by:

$$R_{\text{sec}} = 1 - h(\text{QBER}) - f_{\text{EC}} h(\text{QBER}) \quad (3.1)$$

where $h(\cdot)$ denotes the binary Shannon entropy and f_{EC} is the error correction inefficiency factor.

CHAPTER 4

RESULTS & DISCUSSION

- I. SIMULATION OUTPUTS
- II. BB84 vs B92
- III. PERFORMANCE VALIDATION
- IV. IMPLICATIONS

CHAPTER 5

CONCLUSION & FUTURE WORK

BIBLIOGRAPHY

- [1] D. Kahn, *The Codebreakers: The Story of Secret Writing*. New York: Macmillan, 1967, first edition.
- [2] I. A. Al-Kadi, “Origins of cryptology: The Arab contributions,” *Cryptologia*, vol. 16, no. 2, pp. 97–126, Apr. 1992.
- [3] S. Singh, *The Code Book: The Science of Secrecy from Ancient Egypt to Quantum Cryptography*. New York, NY, USA: Anchor Books/Doubleday, 1999, first published London: Fourth Estate, 1999.
- [4] C. P. Bauer, *Secret History: The Story of Cryptology*, 2nd ed. Boca Raton, FL, USA: Chapman and Hall/CRC, 2021.
- [5] A. Hodges, *Alan Turing: The Enigma*, updated ed. Princeton, NJ, USA: Princeton University Press, 2014.
- [6] H. Williams, “The Navajo Code Talkers: A cryptologic and linguistic perspective,” *Cryptologia*, vol. 24, no. 4, pp. 289–308, 2000.
- [7] C. E. Shannon, “A mathematical theory of cryptography,” Bell Telephone Laboratories, Murray Hill, NJ, USA, Tech. Rep. Memorandum MM 45-110-02, 1945, classified Memorandum, September 1, 1945.
- [8] —, “A mathematical theory of communication,” *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.
- [9] W. Diffie and M. Hellman, “New directions in cryptography,” *IEEE Transactions on Information Theory*, vol. 22, no. 6, pp. 644–654, 1976.
- [10] R. L. Rivest, A. Shamir, and L. Adleman, “A method for obtaining digital signatures and public-key cryptosystems,” *Communications of the ACM*, vol. 21, no. 2, pp. 120–126, Feb. 1978.
- [11] P. W. Shor, “Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer,” *SIAM Journal on Computing*, vol. 26, no. 5, pp. 1484–1509, 1997.
- [12] C. H. Bennett and G. Brassard, “Quantum cryptography: Public key distribution and coin tossing,” *Proceedings of IEEE International Conference on Computers, Systems and Signal Processing*, vol. 175, p. 8, 1984.
- [13] C. H. Bennett, “Quantum cryptography using any two nonorthogonal states,” *Physical Review Letters*, vol. 68, no. 21, pp. 3121–3124, 1992.
- [14] A. K. Ekert, “Quantum cryptography based on bell’s theorem,” *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, 1991.
- [15] D. Gottesman, H.-K. Lo, N. Lütkenhaus, and J. Preskill, “Security of quantum key distribution with imperfect devices,” *Quantum Information & Computation*, vol. 4, no. 5, pp. 325–360, 2004.
- [16] G. Brassard and L. Salvail, “Secret-key reconciliation by public discussion,” in *Advances in Cryptology — EUROCRYPT ’93*, T. Hellese, Ed. Berlin, Heidelberg: Springer Berlin Heidelberg, 1994, pp. 410–423.
- [17] P. W. Shor and J. Preskill, “Simple proof of security of the BB84 quantum key distribution protocol,” *Physical Review Letters*, vol. 85, no. 2, pp. 441–444, Jul 2000.
- [18] A. Boaron, B. Fedrici, R. Houlmann, N. Walenta, H. Liang, J. Liu, W. Zhang, X. Peng, Z. Wang, A. E. Lita, F. Marsili, V. B. Verma, S. W. Nam, M. Fasel, J.-D. Lim, B. Korzh, H. Zbinden, and R. Thew, “Secure quantum key distribution over 421 km of optical fiber,” *Physical Review Letters*, vol. 121, no. 19, p. 190502, Nov 2018.
- [19] H. Zhang, H. Zhu, R. He *et al.*, “Towards global quantum key distribution,” *Nature Reviews Electrical Engineering*, vol. 2, pp. 806–818, 2025. [Online]. Available: <https://www.nature.com/articles/s44287-025-00238-7>
- [20] C. Gidney and M. Ekerå, “How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits,” *Quantum*, vol. 5, p. 433, 2021.
- [21] C. Gidney, “How to factor 2048 bit RSA integers with less than a million noisy qubits,” 2025.
- [22] L. K. Grover, “A fast quantum mechanical algorithm for database search,” in *Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC)*, 1996, pp. 212–219.
- [23] M. Nielsen and I. Chuang, “Quantum computation and quantum information,” Cambridge, 2000.
- [24] National Institute of Standards and Technology, “Mappings of migration to PQC project capabilities to risk framework documents,” U.S. Department of Commerce, Gaithersburg, MD, USA, NIST Cybersecurity White Paper (CSWP) 48, Sep. 2025, initial Public Draft.
- [25] D. Moody, R. Perlner, A. Regenscheid, A. Robinson, and D. Cooper, “Transition to post-quantum cryptography standards,” National Institute of Standards and Technology, NIST Internal Report IR 8547 ipd, Nov. 2024, initial Public Draft.
- [26] V. Scarani, H. Bechmann-Pasquinucci, N. J. Cerf, M. Dušek, N. Lütkenhaus, and M. Peev, “The security of practical quantum key distribution,” *Reviews of Modern Physics*, vol. 81, no. 3, pp. 1301–1350, Sep 2009.
- [27] W. K. Wootters and W. H. Zurek, “A single quantum cannot be cloned,” *Nature*, vol. 299, no. 5886, pp. 802–803, 1982.
- [28] D. Dieks, “Communication by EPR devices,” *Physics Letters A*, vol. 92, no. 6, pp. 271–272, 1982.
- [29] D. Bouwmeester, A. K. Ekert, and A. Zeilinger, “The physics of quantum information : quantum cryptography, quantum teleportation, quantum computation,” London, 2000.

- [30] W. Heisenberg, "Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik," *Zeitschrift für Physik*, vol. 43, no. 3–4, pp. 172–198, 1927.
- [31] A. Einstein, B. Podolsky, and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?" *Physical Review*, vol. 47, pp. 777–780, 1935.
- [32] J. S. Bell, "On the Einstein-Podolsky-Rosen paradox," *Physics Physique Fizika*, vol. 1, no. 3, pp. 195–200, 1964.
- [33] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed experiment to test local hidden-variable theories," *Physical Review Letters*, vol. 23, no. 15, pp. 880–884, 1969.
- [34] A. Aspect, J. Dalibard, and G. Roger, "Experimental test of Bell's inequalities using time-varying analyzers," *Physical Review Letters*, vol. 49, no. 25, pp. 1804–1807, 1982.
- [35] M. Kumar and B. Mondal, "A brief review on Quantum Key Distribution Protocols," *Multimedia Tools and Applications*, 2025. [Online]. Available: <https://link.springer.com/article/10.1007/s11042-024-20535-x>

APPENDIX A

SOURCE CODE

APPENDIX B

ADDITIONAL RESULTS